Can daily OHLCV rank tomorrow’s relative winners?

One public source, two economic hypotheses, one cross-sectional forecasting task

Predict which liquid stocks outperform their universe over the next session.

Hypothesis	Who takes the other side?	Risk that may sustain the return
Momentum: gradual repricing	Investors who respond slowly to information	Reversals in leadership and crowded positioning
Reversal: temporary price pressure	Investors who need immediate execution	Inventory risk and adverse selection

OHLCV can test implications of these mechanisms. It does not observe investor motives.

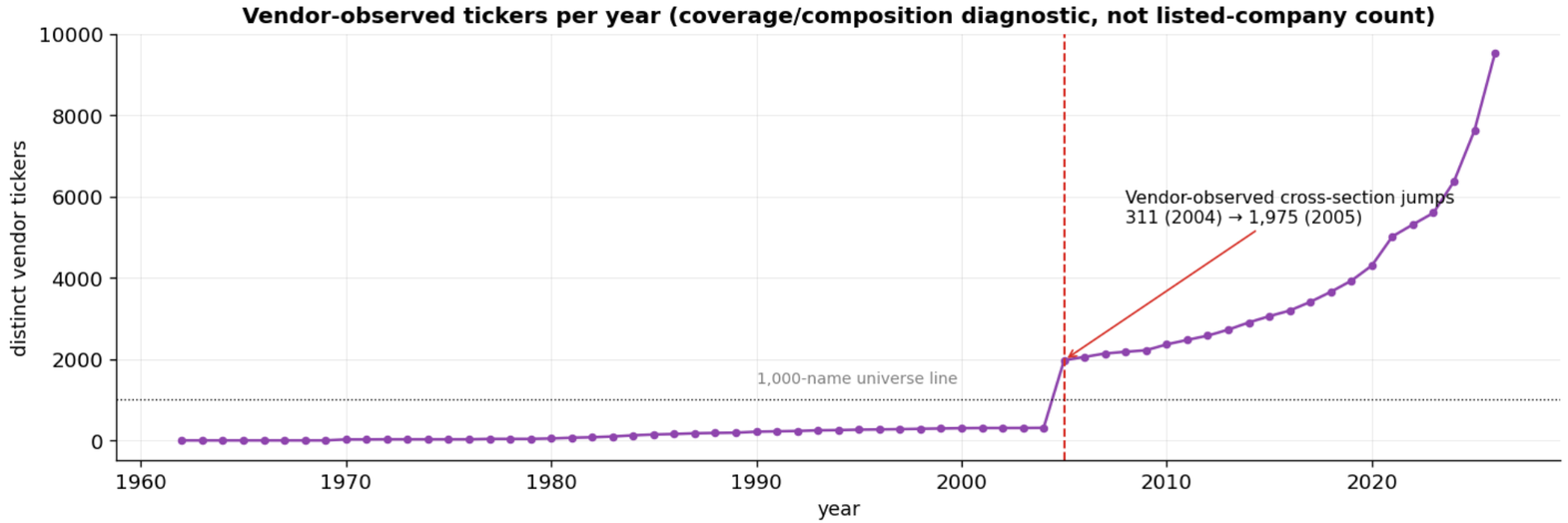
Rishit Belani

DECISION Test the mechanism, then test its economic value.

NEXT Can the source support a credible stock-day panel?

Coverage determines the research sample

Stooq daily open, high, low, close and volume



20.61 million rows

9,530 vendor tickers
1962–September 2026

Research starts in 2005

Coverage expands enough to study roughly 1,000 liquid names.

Ticker / date	Open	High	Low	Close	Volume
AAPL / 1984-09-07	0.099172	0.100390	0.097975	0.099172	99,242,379

DECISION Use broad coverage from 2005 and audit the source's omissions.

NEXT Which rows and instruments should enter the analysis?

Return construction preserves the time axis

Quality exclusions, feature clipping and target treatment have different jobs

$$r^{A}_{i,t} = C_{i,t}/C_{i,t-1} - 1, \qquad \tilde{r}_{i,t} = \mathrm{clip}(r^{A}_{i,t}, -0.5, 0.5)$$

C is close. r^{A} is quality-screened. The feature return is clipped at ±50%.

Problem	Implemented treatment	Reason
Non-equity symbols	Remove 676 whole symbols	Reduce instrument-type noise
Bad / extreme prices	Flag prices and null return events	Preserve interior rows and dates
Missing ticker dates	Require adjacent return endpoints	Prevent multi-day labels
Missing feature history	Require minimum history	No imputation

Targets and P&L retain actual returns after quality exclusions.

DECISION Build features on full histories before filtering the universe.

NEXT Which names and information are available on the signal date?

Universe, information date and target must agree

Lagged liquidity defines selection depth within the available source

$$ADV_{i,t} = \text{median}_{j=1,\ldots,60}(C_{i,t-j}v_{i,t-j})$$

Top 1,000 eligible names by lagged median dollar volume.
60 observed rows, minimum 40 valid values. Mean universe: 990 names.

$$y_{i,t+1} = r^{A}_{i,t+1} - \frac{1}{N_t}\sum_{j \in U^{obs}_t} r^{A}_{j,t+1}$$

U^{obs} contains selected names with a valid adjacent-session outcome.
Subtracting the cross-sectional mean leaves the ranking unchanged.

Feature information	Return window	Timing limitation
Momentum through t–21 Reversal through t–1	Close t to close t+1 Adjacent source sessions	Eligibility also uses current close and volume

DECISION Keep label timing explicit and validate the close-execution convention.

NEXT How do the raw histories become one score per stock?

Momentum and reversal encode different horizons

Every feature uses full observed ticker history before the daily universe filter

Momentum: continuation after skipping the recent month

$$e_{i,t} = \tilde{r}_{i,t} - \bar{\tilde{r}}_t$$

$$M_{i,t} = \sum_{j=21}^{251} e_{i,t-j}$$

231 observed rows, shifted 21, minimum 120 valid returns. A simple-return sum.

Reversal: recent losers receive a higher score

$$V_{i,t}^{(h)} = -\sum_{j=1}^{h} \tilde{r}_{i,t-j}, \qquad h \in \{3, 5, 10\}$$

$$V_{i,t} = \text{mean}_{h \in H_{i,t}} V_{i,t}^{(h)}$$

Average the available complete 3-, 5- and 10-row horizons. Skip the latest return.

Market demeaning removes a common move. Beta and sector exposures remain.
Signal coverage is 97.7% across 5.40 million in-universe stock-days.

DECISION Keep the core horizons fixed when comparing alternatives.

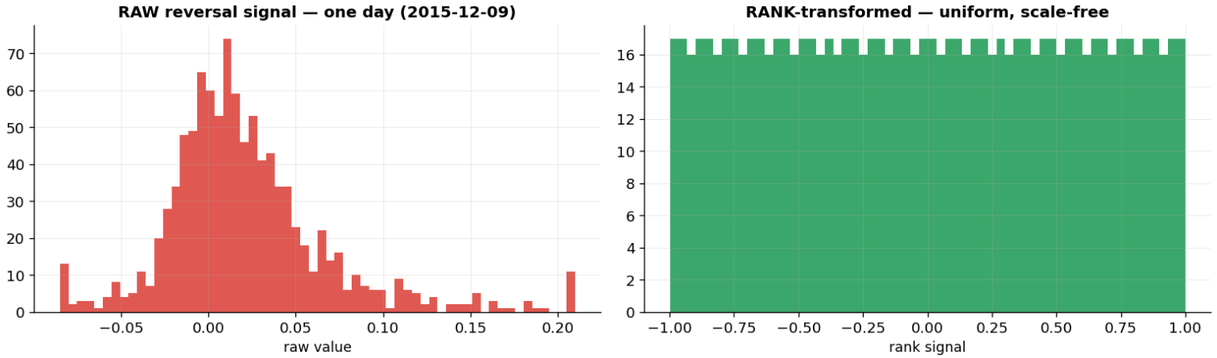
NEXT Do ranks turn these raw features into a useful benchmark?

Ranks create a transparent, complementary benchmark

Equal weighting is the baseline against which added complexity must earn its place

$$R_t(z) = 2\,\mathrm{pctRank}_t(z) - 1$$

$$S_{i,t} = \frac{R_t(M_{i,t}) + R_t(V_{i,t})}{2}, \qquad S^R_{i,t} = R_t(S_{i,t})$$



Signal	Mean IC	IC IR
Momentum	0.0129	1.02
Reversal	0.0071	0.70
Equal blend	0.0132	1.20

$$IC_t = \mathrm{Corr}_i(\mathrm{rank}(S_{i,t}), \mathrm{rank}(y_{i,t+1}))$$

$$IR_{IC} = \sqrt{252}\,\frac{\overline{IC}}{s(IC)}, \qquad \overline{IC}^2 \approx 1.7 \times 10^{-4}$$

Rank IC² is a scale reference, not explained return variance. Signal-rank correlation: –0.017.

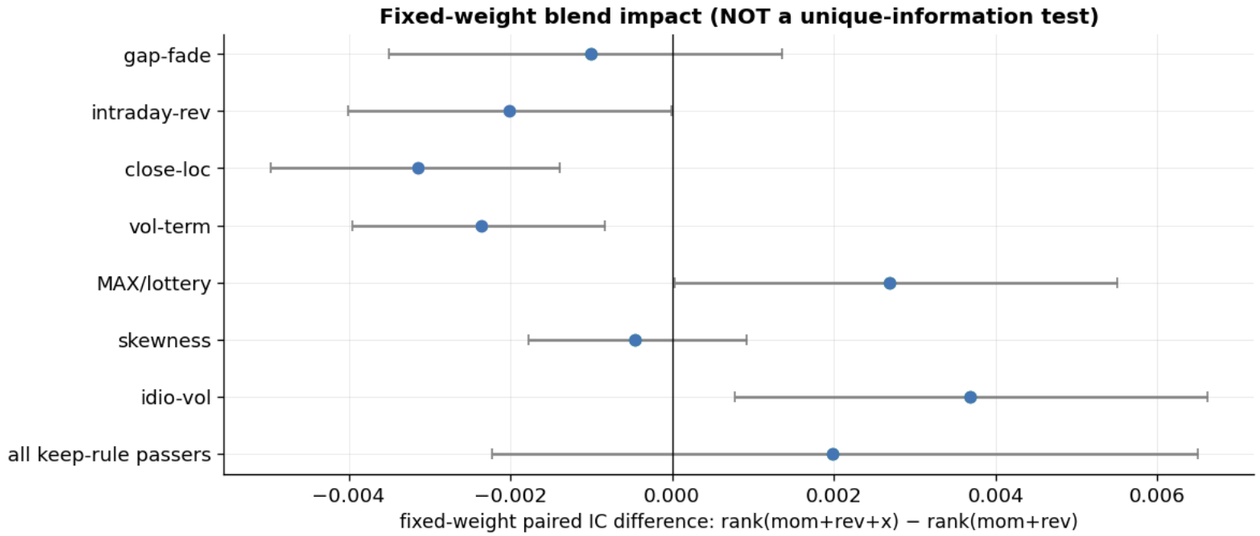
DECISION **Use the blend as a baseline and demand incremental evidence.**

NEXT Which candidate features add information beyond this core?

Low volatility and MAX add related information

Seven candidate features test whether the simple core leaves useful information unexplained

$$z_{i,t}^{\perp} = z_{i,t} - \hat{a}_t - \hat{b}_t R_t(M_{i,t}) - \hat{c}_t R_t(V_{i,t})$$



Candidate	Unique IC	Adjusted p
Low volatility	0.0109	0.0005
Low MAX	0.0091	0.0005
Low skewness	0.0035	0.189

Seven-candidate joint block bootstrap.
Low-vol / MAX rank correlation: 0.676.

DECISION Treat low volatility and MAX as one related signal family.

NEXT Can a fitted model use these state characteristics effectively?

The ML design separates weights, state and nonlinearity

Annual expanding fits predict ranks on common evaluation rows and dates

State features before percentile ranking

$$\sigma_{i,t} = \mathrm{sd}(\tilde{r}_{i,t-1:t-20}) \qquad a_{i,t} = |\tilde{r}_{i,t-1}|, \qquad \ell_{i,t} = \log(ADV_{i,t})$$

Volatility, prior absolute move and log dollar volume. ADV is a size proxy.

$$q_{i,t+1} = \mathrm{pctRank}_t(y_{i,t+1})$$

$$\widehat{\beta} = \arg\min_{\beta_0,\beta} \left\{ \sum_{(i,t)\in train} (q_{i,t+1} - \beta_0 - X_{i,t}^{\top}\beta)^2 + \lambda\|\beta\|_2^2 \right\}$$

$$\hat{q}(X) = f_0(X) + \eta \sum_{b=1}^{B} f_b(X)$$

Core Ridge: learned weights.
Conditioned Ridge: state effects.
Boosting: nonlinear interactions.

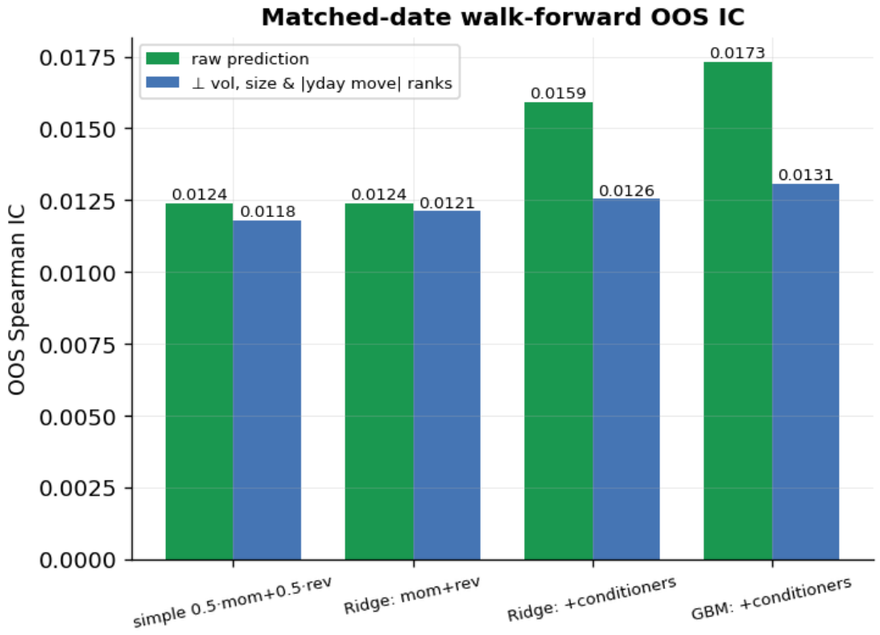
Three-year initial history. Expanding annual refits. No random train/test split.
Boundary labels are not explicitly purged, which remains a validation limitation.

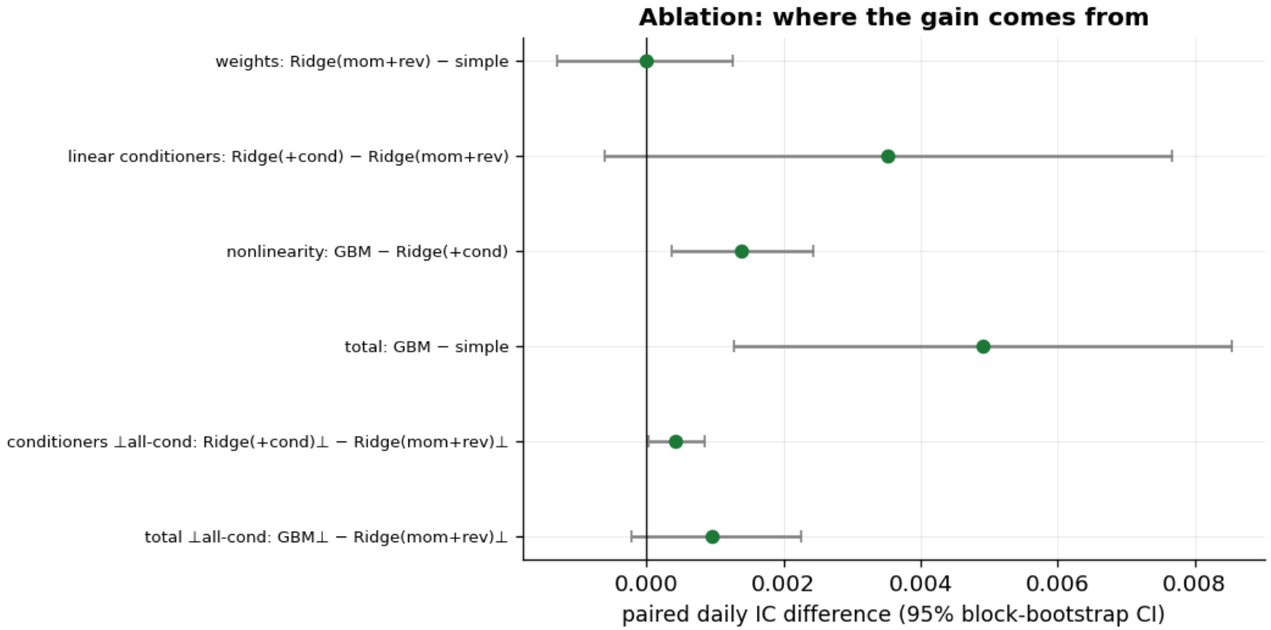
DECISION Compare the models on the same target and historical test dates.

NEXT Does the gain survive uncertainty and exposure diagnostics?

Raw prediction gains shrink after removing state exposures

4,692 common evaluation dates let the ablation compare like with like





Raw IC: blend 0.0124, core Ridge 0.0124, conditioned Ridge 0.0159, GBM 0.0173.

GBM gain over the blend: +0.00491, 95% block CI [0.00128, 0.00854].

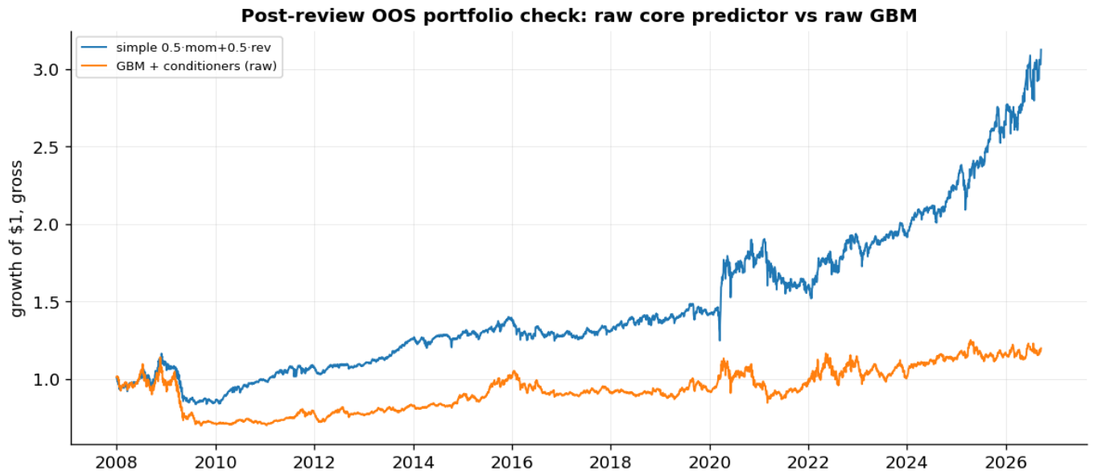
DECISION Preserve raw gains while attributing the exposures behind them.

NEXT Does the higher rank IC improve the same portfolio construction?

Higher rank IC does not improve this portfolio

IC uses the full cross-section. The portfolio selects tails and applies volatility weights.

$$w^{L}_{i,t} = \frac{0.5/\sigma_{i,t}}{\sum_{j \in L_t} 1/\sigma_{j,t}}, \qquad \sum_i w_{i,t} = 0, \quad \sum_i |w_{i,t}| = 1$$



Metric	Equal blend	GBM
Gross Sharpe	0.566	0.141
Max drawdown	–27.99%	–38.65%
Turnover/day	0.621	0.702
Break-even bps	4.38	0.99

Next diagnostic: equal-weight deciles, then risk sizing.

$$P_{t+1} = \sum_i w_{i,t} r^{A}_{i,t+1}, \qquad SR = \sqrt{252}\,\frac{\bar{P}}{s(P)}$$

DECISION **The gross gap precedes costs. Diagnose tails and sizing before promoting GBM.**

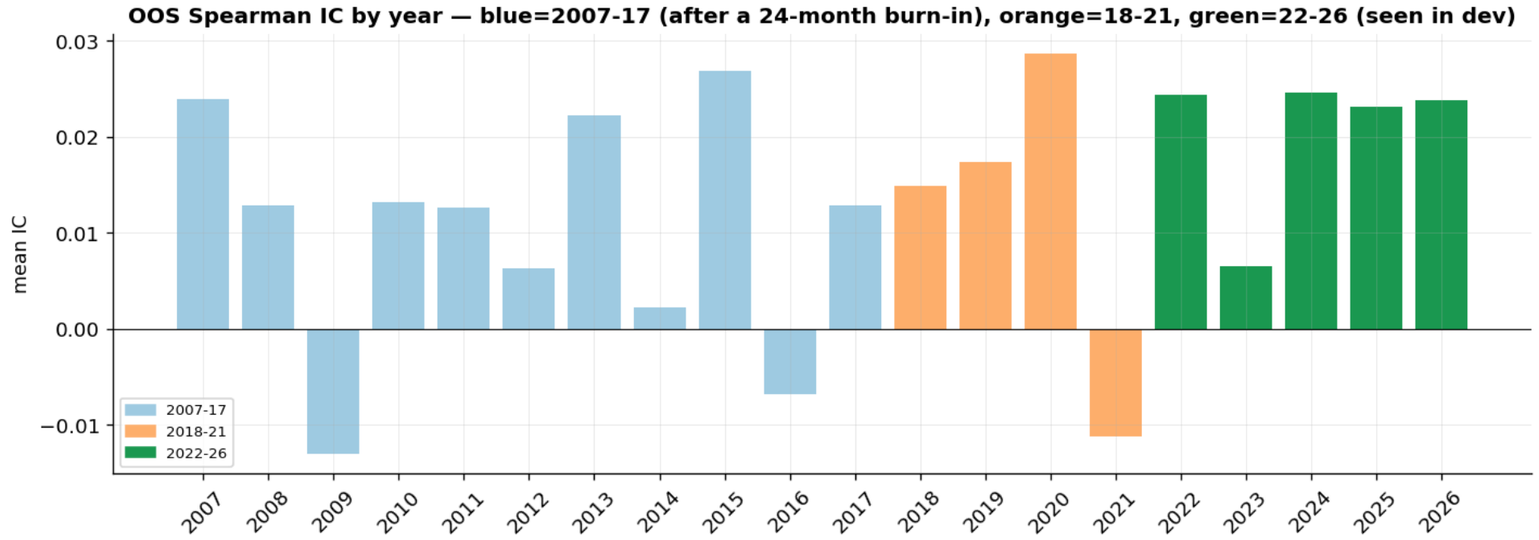
NEXT How credible is the historical evidence under chronological evaluation?

Positive walk-forward IC remains development evidence

Monthly core weights use past IC after a 24-month burn-in

$$w_{k,m} = \frac{[\overline{IC}_{k,<m}]_+}{\sum_{\ell \in \{M,V\}} [\overline{IC}_{\ell,<m}]_+}$$

$$CI_{95\%} = \left[Q_{.025}(\overline{IC}^*),\ Q_{.975}(\overline{IC}^*)\right]$$



Mean IC: 0.0131

95% CI
[0.0083, 0.0178]
4,943 dates

2007–17: 0.0103
2018–21: 0.0125
2022–26: 0.0203

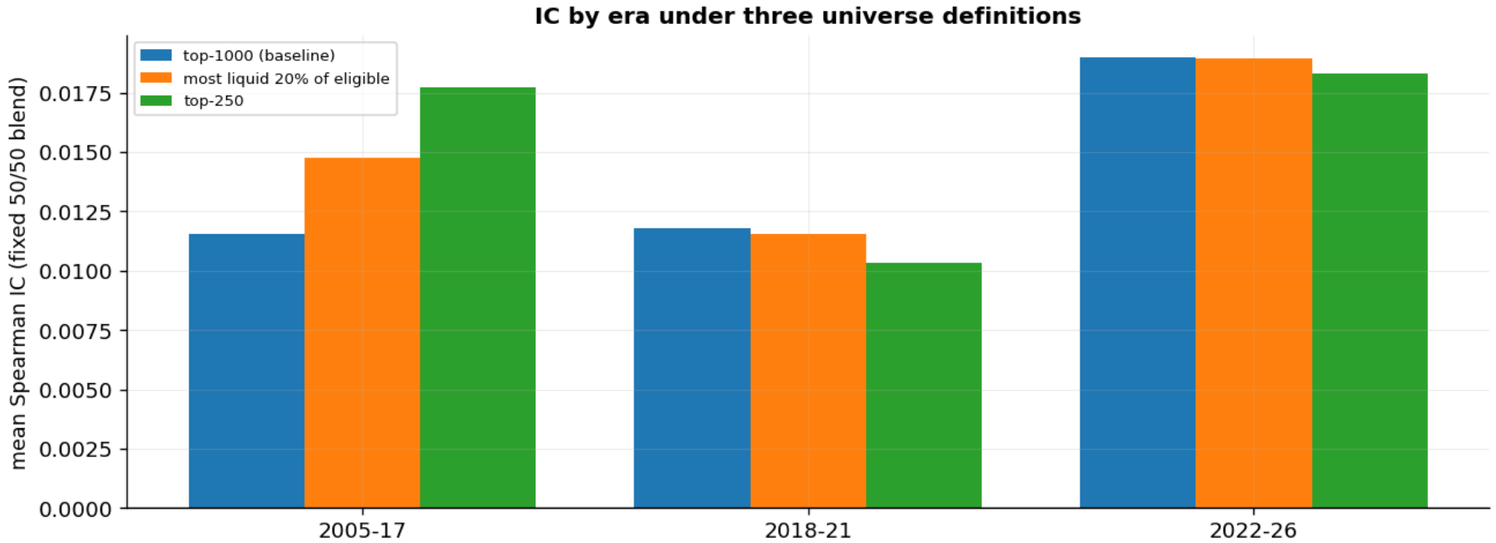
Historical development evidence. Recent years were inspected. No untouched holdout.

DECISION Interpret uncertainty conditional on this data and research design.

NEXT Does stronger recent IC partly reflect a changing universe?

Recent strength depends on universe composition

The signal stays fixed while the liquidity selection rule changes



Universe	Recent – early IC
Top 1,000	+0.0074
Top 20%	+0.0042
Top 250	+0.0006

Top 1,000 selects 54% of eligible names in 2005 and 14% in 2026.
The 1,000th name's median dollar volume rises from $2.1M to $89.4M.

Same-source checks retain positive IC but cannot recover missing delisted stocks.
Confidence remains conditional on historical coverage and security identity.

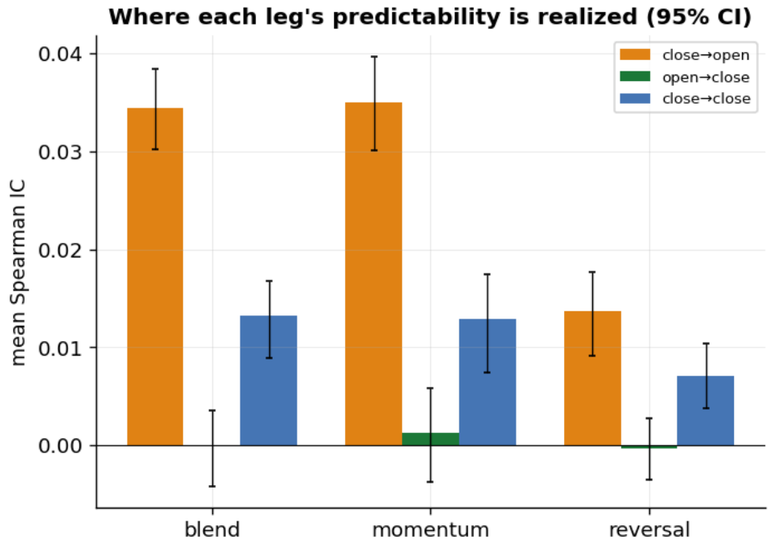
DECISION Treat recent strength as sensitive to universe composition.

NEXT In which holding period does the predictive information appear?

Predictability concentrates in the overnight return

The same score ranks two components of the adjacent-session outcome

$$r^{ON}_{i,t+1} = O_{i,t+1}/C_{i,t} - 1, \qquad r^{ID}_{i,t+1} = C_{i,t+1}/O_{i,t+1} - 1$$



Signal	Overnight IC	Intraday IC
Momentum	0.0350	0.0013
Reversal	0.0137	–0.0003
Blend	0.0344	–0.0001

$$1 + r^{CC}_{i,t+1} = (1 + r^{ON}_{i,t+1})(1 + r^{ID}_{i,t+1})$$

DECISION **Treat the overnight window as an implementation hypothesis.**

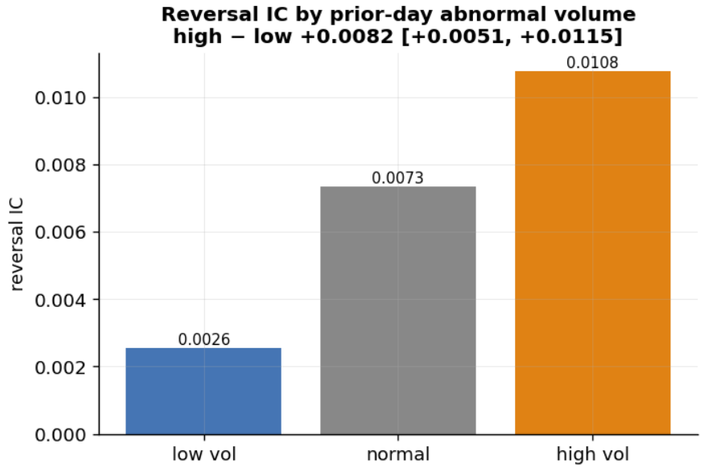
NEXT Do volume and market conditions support the reversal mechanism?

Volume and dispersion are consistent with conditional reversal

Demand for immediacy and the risk of supplying liquidity motivate the tests

$$AV_{i,t} = \frac{DV_{i,t-1}}{\text{median}(DV_{i,t-2:t-21})}$$

$$D_{t-1} = \text{sd}_{i \in U_{t-1}}(r^{A}_{i,t-1})$$



Abnormal volume

High – low IC: +0.0082
95% CI [0.0051, 0.0115]

Past-only dispersion states

High – low IC: +0.0105
95% CI [0.0012, 0.0194]

Causal test needs signed flow, auction imbalances and quotes, with controls for news.

DECISION Retain conditional evidence while keeping causality unresolved.

NEXT Can the measured edge cover the trading required?

Turnover turns a small gross edge into a cost budget

Continuous and overnight books use different trading conventions

Holding window	Gross Sharpe	Turnover/day	Break-even bps/$	Net Sharpe at 2 bps/$
Continuous close–close	0.539	0.566	4.57	0.303
Close–open, flatten daily	1.543	2.000	1.83	–0.143

$$\tau_t^{CC} \approx \sum_i |w_{i,t} - w_{i,t-1}|, \qquad \tau_t^{ON} = 2\sum_i |w_{i,t}|$$

$$P_{t+1}^{net} = P_{t+1}^{gross} - c\tau_t, \qquad c_{BE} = 10^4 \frac{\bar{P}^{gross}}{\bar{\tau}} \text{ bps}$$

One basis point per dollar traded consumes about 1.43% per year in the continuous book.

DECISION **Compare expected all-in cost with the measured break-even budget.**

NEXT Which single experiment would most change confidence in this signal?

The next experiment is independent replication

Priority: test whether the frozen core survives a cleaner historical equity universe

Test element	Precommitted specification	Interpretation
Data	Common stocks, delistings and verified sessions	Independent source
Freeze	Core features, universe rules, weights and evaluation	Before opening results
Pass	Mean IC > 0.005 and 95% block CI above zero	Increase confidence
Fail	The gate fails or the effect reverses	Record non-replication

Failure stops advancement of this specification. Do not retune on the replication sample.

Historical replication tests source bias. A locked future holdout tests generalization.

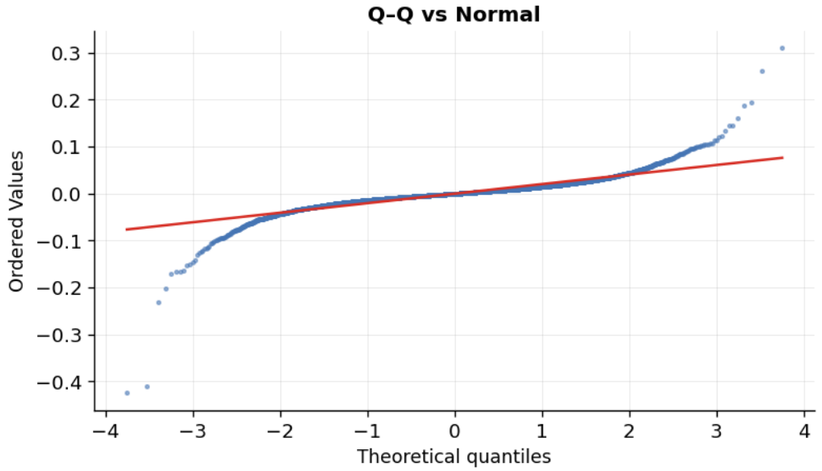
DECISION Resolve data and universe uncertainty before adding model complexity.

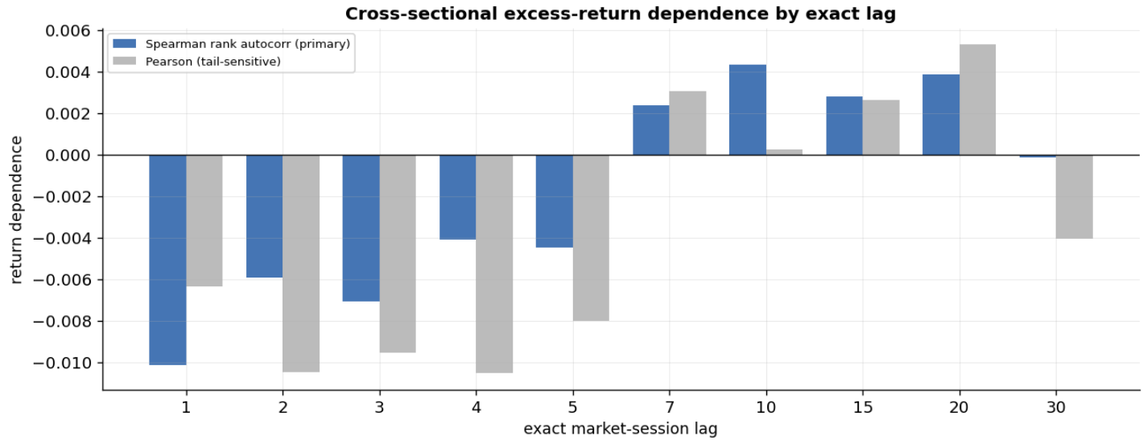
DISCUSSION What evidence would change this research decision?

Heavy tails and changing dependence shape the method

BACKUP: The post-clean target still contains substantial noise and regime variation

$$\hat{\kappa} = \frac{m_4}{m_2^2} - 3 = 51.7, \qquad m_k = \frac{1}{n}\sum_{a=1}^{n}(y_a - \bar{y})^k$$





Excess kurtosis: 51.7
Target standard deviation: 2.27%

Rank dependence is negative at lags 1–5.
Daily dispersion ranges from 0.84% to 9.86%.

DECISION Use ranks, chronological evaluation and uncertainty over dates.

NEXT How much information does the transparent core contain?

Seven raw candidate features test additional mechanisms

BACKUP: All formulas use lagged OHLCV and become daily cross-sectional ranks

Short-horizon price pressure

Gap fade

$$G_{i,t} = -(O_{i,t-1}/C_{i,t-2} - 1)$$

Intraday fade

$$I_{i,t} = -(C_{i,t-1}/O_{i,t-1} - 1)$$

Close location

$$CL_{i,t} = -\frac{C_{i,t-1} - L_{i,t-1}}{H_{i,t-1} - L_{i,t-1}}$$

Volatility term structure

$$VT_{i,t} = -\frac{\mathrm{sd}(\tilde{r}_{i,t-1:t-5})}{\mathrm{sd}(\tilde{r}_{i,t-1:t-60})}$$

Volatility and lottery preference

Low MAX

$$MAX_{i,t} = -\max_{j=1,\ldots,21} \tilde{r}_{i,t-j}$$

Low skewness

$$SK_{i,t} = -\mathrm{skew}(\tilde{r}_{i,t-1:t-60})$$

Low market-demeaned volatility

$$LV_{i,t} = -\mathrm{sd}(e_{i,t-1:t-60})$$

Raw recipes use stated signs.
Augmentation sets signs from prior-year IC.

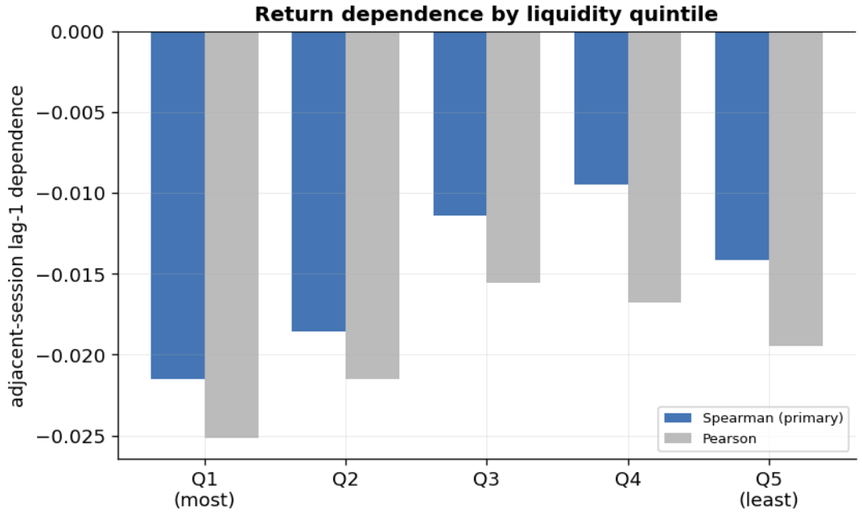
DECISION Screen the stated sign, then test conditional information.

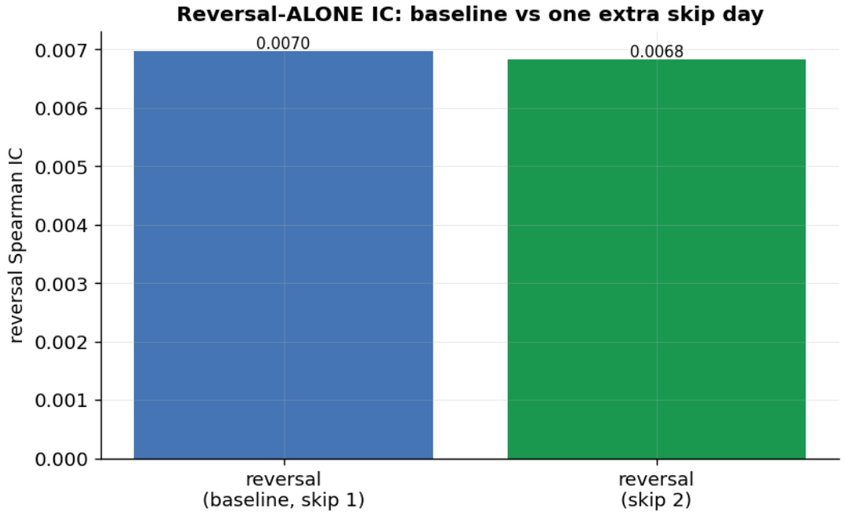
NEXT Which candidates survive a test against the existing core?

Delayed reversal survives one additional skip

BACKUP: The test sacrifices the strongest immediate lag to examine persistence

$$V_{i,t}^{(h,k)} = -\sum_{j=k}^{k+h-1} \tilde{r}_{i,t-j}, \qquad k \in \{0,1,2\}$$





Skip 0: IC 0.0109 **Skip 1: IC 0.0070** **Skip 2: IC 0.0068**

DECISION Retain delayed reversal and validate executable timing separately.

NEXT Is the predictive information realized overnight or intraday?

Execution validation connects forecasts to realized value

BACKUP: Pre-trade feasibility and post-trade attribution use an explicit benchmark

$$IS_{bps}^{fill} = 10^4 \, d \, \frac{P_{fill} - P_{benchmark}}{P_{benchmark}} + \text{fees}_{bps}$$

d = +1 for a buy and –1 for a sell. Completed-fill cost uses a stated benchmark.

Stage	Question to answer	Data required next
Pre-trade	Can expected costs fit within the edge budget?	Spread, auction liquidity, participation and borrow
Post-trade	Where did realized fills differ from the benchmark?	Decision / arrival prices, fills, fees and timestamps
Intra- / multi-day	Does the execution path preserve alpha after risk and delay?	Intraday liquidity, volatility and alpha decay

23 missing position-days. Adverse stress lowers gross Sharpe from 0.539 to 0.487.

DECISION Use the cost budget to prioritize execution data and attribution.

NEXT Which next test could most change the research conclusion?